

Problem 1

Find the complex numbers corresponding to the points:

$$1^\circ(0, 1); \quad 2^\circ(1, 0); \quad 3^\circ(0, -1); \quad 4^\circ(-1, 0); \quad 5^\circ(-2, 3).$$

Solution

We will convert each point (x, y) in the Cartesian plane to its corresponding complex number $z = x + iy$.

Recall: Complex Number Representation

A point (x, y) in the Cartesian plane (complex plane) corresponds to the complex number:

$$z = x + iy$$

where:

- x is the real part: $\text{Re}(z) = x$
- y is the imaginary part: $\text{Im}(z) = y$
- i is the imaginary unit with $i^2 = -1$

Part 1^o: Point $(0, 1)$

Conversion

For the point $(0, 1)$:

- $x = 0$ (real part)
- $y = 1$ (imaginary part)

The corresponding complex number is:

$$z = 0 + i \cdot 1 = i$$

Answer

$$\boxed{z = i}$$

This is a purely imaginary number on the positive imaginary axis.

Part 2^o: Point $(1, 0)$

Conversion

For the point $(1, 0)$:

- $x = 1$ (real part)
- $y = 0$ (imaginary part)

The corresponding complex number is:

$$z = 1 + i \cdot 0 = 1$$

Answer

$$z = 1$$

This is a purely real number on the positive real axis.

Part 3°: Point $(0, -1)$

Conversion

For the point $(0, -1)$:

- $x = 0$ (real part)
- $y = -1$ (imaginary part)

The corresponding complex number is:

$$z = 0 + i \cdot (-1) = -i$$

Answer

$$z = -i$$

This is a purely imaginary number on the negative imaginary axis.

Part 4°: Point $(-1, 0)$

Conversion

For the point $(-1, 0)$:

- $x = -1$ (real part)
- $y = 0$ (imaginary part)

The corresponding complex number is:

$$z = -1 + i \cdot 0 = -1$$

Answer

$$z = -1$$

This is a purely real number on the negative real axis.

Part 5°: Point $(-2, 3)$

Conversion

For the point $(-2, 3)$:

- $x = -2$ (real part)
- $y = 3$ (imaginary part)

The corresponding complex number is:

$$z = -2 + i \cdot 3 = -2 + 3i$$

Answer

$$z = -2 + 3i$$

This is a complex number in the second quadrant (negative real part, positive imaginary part).

Summary Table

Point (x, y)	Complex Number z	Real Part	Imaginary Part
$(0, 1)$	i	0	1
$(1, 0)$	1	1	0
$(0, -1)$	$-i$	0	-1
$(-1, 0)$	-1	-1	0
$(-2, 3)$	$-2 + 3i$	-2	3

Additional Properties

Modulus (Magnitude)

For each complex number $z = x + iy$, the modulus is:

$$|z| = \sqrt{x^2 + y^2}$$

- $|i| = \sqrt{0^2 + 1^2} = 1$
- $|1| = \sqrt{1^2 + 0^2} = 1$
- $|-i| = \sqrt{0^2 + (-1)^2} = 1$
- $|-1| = \sqrt{(-1)^2 + 0^2} = 1$
- $|-2 + 3i| = \sqrt{(-2)^2 + 3^2} = \sqrt{4 + 9} = \sqrt{13}$

Argument (Angle)

The first four points lie on the unit circle and correspond to special angles:

- i : $\arg(i) = \frac{\pi}{2}$ (90°)
- 1 : $\arg(1) = 0$ (0°)
- $-i$: $\arg(-i) = -\frac{\pi}{2}$ or $\frac{3\pi}{2}$ (90° or 270°)
- -1 : $\arg(-1) = \pi$ (180°)
- $-2 + 3i$: $\arg(-2 + 3i) = \pi - \arctan\left(\frac{3}{2}\right)$ (in quadrant II)

Final Answer

1°	$(0, 1) \rightarrow z = i$
2°	$(1, 0) \rightarrow z = 1$
3°	$(0, -1) \rightarrow z = -i$
4°	$(-1, 0) \rightarrow z = -1$
5°	$(-2, 3) \rightarrow z = -2 + 3i$

General Formula: For any point (x, y) in the complex plane, the corresponding complex number is:

$$z = x + iy$$

Note: The first four complex numbers $(i, 1, -i, -1)$ are the fourth roots of unity multiplied by appropriate powers, forming the vertices of a square on the unit circle.